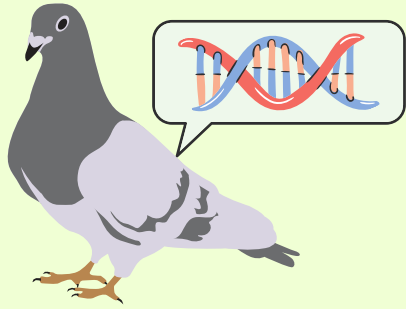


CRISPR AND THE REVIVAL OF THE ICONIC PASSENGER PIGEON

De-extinction is an emerging field of study that has the potential to restore lost biodiversity and provide novel approaches to reviving extinct species [1]. Once a highly nomadic bird, the passenger pigeon went extinct in 1914. Its communal lifestyle made mass hunting of the species simple because of their unpredictable travel patterns and constant search for food [2,3]. Genetic engineering has advanced significantly thanks to genome editing, which enables researchers to make precise modifications to an organism's DNA via CRISPR/Cas9 [4].

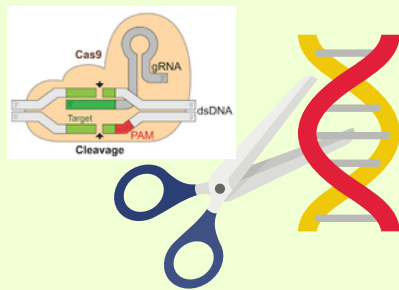


1. EXTRACTION OF DNA

Obtain and manage fragmented DNA from conserved passenger pigeon specimens [5].

2. SEQUENCING AND ANALYSIS:

Examine the retrieved DNA's genetic code and contrast it with those of related extant species (Band-tailed pigeon) [6].

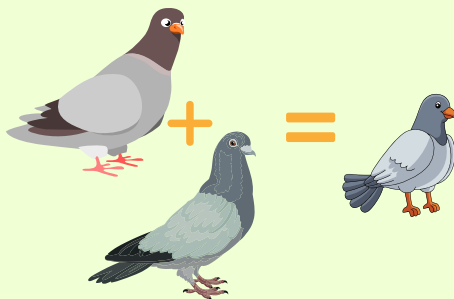


3. CRISPR-CAS9

Gene editing is used to modify the DNA of a similar living species to match the genetic traits of the passenger pigeon [7].

4. EMBRYO DEVELOPMENT

Insert the altered genes into live egg embryos to create species with passenger pigeon characteristics [8].



5. BREEDING AND RAISING

In regulated habitats, breed and raise these genetically altered Pigeons [8].

6. Release and monitoring:

These birds might be released into the wild, and their adaption and influence on ecosystems would be monitored [9].



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